The Euler-transform candidate need not optimize the generating-function correlation

Principia Math In Progress

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MathDB 356222. Full disproof of Conjecture 5.93.

Task 1: the intended optimization problem

For an invertible lower-triangular kernel D, Schmidt [1, Notation 5.85 and Question 5.86, pp. 79–80] considers

$$\mathrm{Corr}(C,D)=\sum_{n\ge 1}\frac{1}{n}\frac{\sum_{k=1}^{n}|c_kD^{-1}(n,k)|}{\left(\sum_{k=1}^{n}c_k^2\right)^{1/2}\left(\sum_{k=1}^{n}D^{-1}(n,k)^2\right)^{1/2}},\qquad c_k=[q^k]C(q). \tag{1}$$

The stated optimization ranges over rational formal power series with $C(0)\neq 0$, or normalized integer series with $C(0)=1$, and requires a finite score. Conjecture 5.93 [1, p. 87] proposes the Euler-transform expression

$$C_D(q)=\exp\left(-\sum_{n\ge 1}\frac{q^n}{n}\sum_{k=1}^{n}kD(n,k)\right) \tag{2}$$

as an optimal selection. We disprove its actual optimality: an admissible kernel gives a well-defined candidate with score strictly between the scores of two admissible integer polynomials. Thus the counterexample is independent of the additional product formula printed alongside (2).

Theorem 1. *Let $D(n,k)=\delta_{nk}$, so $D^{-1}=D$, and write*

$$E(q)=\exp\left(-\frac{q}{1-q}\right),\qquad P_N(q)=1+\sum_{k=1}^{N}2^{k-1}q^k.$$

Every P_N has a finite score, and

$$\mathrm{Corr}(P_N,D)>\frac{\sqrt{3}}{2}H_N,\qquad H_N=\sum_{n=1}^{N}\frac{1}{n}. \tag{3}$$

Consequently, finite scores are unbounded even on normalized integer polynomials, so no finite global maximum exists. The proposed candidate (2) is E, belongs to $\mathbb{Q}[[q]]$, and satisfies

$$1=\mathrm{Corr}(1+q,D)<\mathrm{Corr}(E,D)<2<\frac{49\sqrt{3}}{40}<\mathrm{Corr}(P_6,D). \tag{4}$$

In particular, E is neither a maximum nor a minimum.

Proof. The identity kernel is lower triangular, invertible, integral, and has diagonal entries one. The inner sum in (2) is n, giving $C_D = E$. Its constant and linear coefficients are $1, -1$; its coefficients are rational by the formal exponential expansion. For any real coefficients with $c_1 \neq 0$, (1) becomes

$$F(C) := \operatorname{Corr}(C, D) = \sum_{n\geq 1} \frac{|c_n|}{n\left(\sum_{k=1}^{n} c_k^2\right)^{1/2}}. \tag{5}$$

All denominators used below are positive, including those in the zero tail of a polynomial. The first summand is one.

For P_N, only the first N summands can be nonzero and

$$\sum_{k=1}^{n} c_k^2 = \frac{4^n - 1}{3}, \qquad \frac{c_n}{\left(\sum_{k=1}^{n} c_k^2\right)^{1/2}} = \frac{\sqrt{3}}{2\sqrt{1 - 4^{-n}}} > \frac{\sqrt{3}}{2} \quad (1 \leq n \leq N).$$

This proves (3). Since the harmonic sums are unbounded, it also proves the nonexistence of a finite maximum, with every individual competitor satisfying the source's finiteness requirement.

It remains to prove the finite upper bound for E, without assuming any asymptotic formula for its coefficients. For $|z| < 1$,

$$\Re \frac{z}{1 - z} = -\frac{1}{2} + \frac{1 - |z|^2}{2|1 - z|^2} > -\frac{1}{2}, \qquad |E(z)|^2 < e. \tag{6}$$

Write $E(z) = \sum_{n\geq 0} e_n z^n$. For $0 < t < 1$, uniform convergence of the power series on $|z| = t$, followed by orthogonality of exponentials, gives

$$\sum_{n\geq 0} e_n^2 t^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |E(te^{i\theta})|^2 \, d\theta \leq e.$$

Letting $t \uparrow 1$, monotone convergence and $e_0 = 1$ show $\sum_{n\geq 1} e_n^2 \leq e - 1 < 2$. Here $e < 3$ follows directly from its factorial series. Also

$$\sum_{n\geq 1} \frac{1}{n^2} < 1 + \sum_{n\geq 2} \frac{1}{n(n-1)} = 2.$$

Since $e_1 = -1$, every coefficient norm in (5) is at least one. Cauchy–Schwarz therefore proves absolute convergence and

$$F(E) \leq \sum_{n\geq 1} \frac{|e_n|}{n} \leq \left(\sum_{n\geq 1} e_n^2\right)^{1/2} \left(\sum_{n\geq 1} n^{-2}\right)^{1/2} < 2. \tag{7}$$

The expansion $E(q) = 1 - q - \frac{1}{2}q^2 + O(q^3)$ implies $F(E) \geq 1 + 1/(2\sqrt{5}) > 1$. Finally, $H_6 = 49/20$, and $(49\sqrt{3})^2 = 7203 > 6400 = 80^2$. Equations (3) and (7) give all the strict inequalities in (4). □

A sharp finite-degree statement

Proposition 1. *For $N \geq 1$, the supremum of (5) over normalized integer polynomials of degree at most N, with $c_1 \neq 0$, is H_N. For $N \geq 2$ the supremum is not attained. The minimum is one, attained precisely when all $c_n = 0$ for $n \geq 2$.*

Proof. Each summand in (5) is at most $1/n$. For $n = 2$ it is strictly less than $1/2$ because $c_1 \neq 0$, even if $c_2 = 0$. For integers $R \geq 2$, the normalized integer polynomial $1 + \sum_{k=1}^{N} R^{k-1} q^k$ has n-th coefficient ratio

$$\frac{R^{n-1}}{\sqrt{1 + R^2 + \cdots + R^{2n-2}}} = \sqrt{\frac{1 - R^{-2}}{1 - R^{-2n}}} \longrightarrow 1.$$

The finite sum tends to H_N. The minimum characterization follows from the first summand being one and all others nonnegative. □

Scope and attribution

The source explicitly allows all invertible lower-triangular kernels and finite-score rational OGFs; the identity kernel and all competitors above satisfy those conditions. No divergent-score series is used as a competitor. The printed infinite-product expression has a separate formal convergence issue; neither that issue nor a change in normalization is the basis of this disproof. The power series (2) is actually outperformed by a normalized integer polynomial. The specialized Lambert-series optimization problem is not resolved by this identity-kernel example. The common obstruction to general maximizer existence also bears on neighboring proposals, but only the selected Conjecture 5.93 receives a campaign credit.

Tasks 2–5: proposed mathematical follow-ups

Task 2: Characterize invertible lower-triangular kernels D for which the finite values of (1) are uniformly bounded over normalized integer polynomials with nonzero linear coefficient.

Task 3: For the Lambert kernel $D(n, k) = \mathbf{1}_{k|n}$, determine whether a finite maximum exists over a precisely specified class of rational OGFs, and whether $(q; q)_\infty$ attains it.

Task 4: With fixed degree N, $c_1 = 1$, and coefficient bounds $|c_k| \leq B$, determine the exact maximizers of the identity-kernel score and their behavior as N or B grows.

Task 5: Replace $1/n$ in (1) by a positive summable weight sequence, and find coefficient-growth constraints giving both existence of a maximizer and a useful Euler-transform characterization.

References

[1] M. D. Schmidt, *Factorization theorems and canonical representations for generating functions of special sums*, Georgia Institute of Technology doctoral dissertation, August 2022, Chapter 5, pp. 78–89; arXiv:2209.12287v1. Official final dissertation: Georgia Tech repository.